A smooth-conical partial answer for smooth singular cochains on real algebraic sets

Autonomous proof attempt for arXiv:2208.11431

11 August 2026

Abstract

The source asks whether the coaugmentation from the constant sheaf to the sheafification of smooth singular cochains on the real spectrum of every finitely generated real algebra is a quasi-isomorphism. We prove the desired conclusion whenever the underlying predifferentiable space is locally smoothly contractible. This gives a concrete positive answer for locally smooth-conical algebraic sets, in particular for isolated positively weighted-homogeneous singularities and for normal-crossing algebraic sets. The argument is stalkwise and uses the smooth singular prism operator. We also isolate why the known Lipschitz contraction does not settle the general question.

1 The exact question

Let B be a finitely generated real algebra and let $X = \operatorname{spec} B$ carry the predifferentiable structure induced by a distinguished algebraic embedding into Euclidean space. Write $C^q_{\mathrm{sm}}(U)$ for the dual of the real vector space of smooth singular q-chains in U, and let

$$^+C^\bullet_{\mathrm{sm},X}$$

be the sheafification of this presheaf complex. The source asks, immediately after its Diagram (7), whether

$$\epsilon : \mathbb{R}_X[0] \longrightarrow {}^+C^\bullet_{\mathrm{sm},X} \tag{1}$$

is a quasi-isomorphism, equivalently whether the induced map on hypercohomology is an isomorphism. The source proves the Lipschitz analogue using local Lipschitz contractibility of algebraic sets, but does not know the smooth analogue.

2 A local criterion

Call a predifferentiable space X *locally smoothly contractible* if, for every $x \in X$ and every neighborhood U of x, there are a neighborhood $V \subset U$ of x and a smooth map

$$H : V \times [0,1] \longrightarrow U$$

such that $H(v,0) = v$ and $H(v,1) = x$.

Theorem 1. *If X is locally smoothly contractible, then the coaugmentation* (1) *is a quasi-isomorphism. Consequently it induces isomorphisms*

$$\mathbb{H}^k(X, \mathbb{R}_X[0]) \;\cong\; \mathbb{H}^k(X, {}^+C^\bullet_{\mathrm{sm},X}) \qquad (k \geq 0).$$

Proof. The stalk of a sheafification equals the stalk of the original presheaf. Moreover, filtered colimits of real vector spaces are exact. We may therefore compute the cohomology sheaves of ${}^{+}C^{\bullet}_{\mathrm{sm},X}$ by taking the cohomology of its stalk complexes.

Fix $x \in X$. A germ in the stalk is represented by a smooth singular cochain on some neighborhood U of x. After shrinking U if necessary, a cocycle germ is represented by an actual cocycle. Choose V and H as in the definition. The standard prism decomposition of $\Delta^q \times [0,1]$ is affine on finitely many simplices. Composing its pieces with $H \circ (\sigma \times \mathrm{id})$ therefore defines the usual smooth singular chain homotopy between the inclusion $V \hookrightarrow U$ and the constant map $V \to \{x\} \hookrightarrow U$. Thus restriction from U to V factors on smooth singular cohomology through the cohomology of a point. It is zero in positive degrees. In degree zero the same argument, or smooth path connectedness of V supplied by H, identifies the kernel of the coboundary with the constant functions.

It follows that every positive-degree cohomology germ vanishes and that the degree-zero cohomology sheaf is $\mathbb{R}_X$. Hence (1) is a quasi-isomorphism. Hypercohomology preserves quasi-isomorphisms. □

3 Weighted cones

For positive integers $w_1, \dots, w_r$, write

$$\delta_t(z_1, \dots, z_r) = (t^{w_1} z_1, \dots, t^{w_r} z_r), \qquad 0 \le t \le 1.$$

A subset $C \subset \mathbb{R}^r$ is a positive weighted cone if $\delta_t(C) \subset C$ for all $t \in [0,1]$. The map $(z,t) \mapsto \delta_t z$ is polynomial and hence smooth, including at $t = 0$.

Say that X is *locally smooth-conical* if every point has a predifferentiable neighborhood smoothly isomorphic to a relatively open neighborhood of $(0,0)$ in $M \times C$, where M is a smooth manifold chart centered at 0 and C is a positive weighted cone.

Corollary 2. *For every locally smooth-conical real algebraic set X, the coaugmentation* (1) *is a quasi-isomorphism and the questioned hypercohomology map is an isomorphism in every degree.*

Proof. Shrink the manifold chart to a ball and contract it linearly to its center. Simultaneously contract C by δ_t. On a sufficiently small product neighborhood this is a smooth contraction. Transfer it through the local smooth isomorphism and apply Theorem 1. □

Corollary 3. *The conclusion holds in each of the following cases.*

1. *X has isolated singularities and, at every singular point, its germ is smoothly equivalent to a positive weighted cone. In particular this covers isolated positively weighted-homogeneous real algebraic singularities.*

2. *X has normal-crossing local models (more generally, local models that are products of a manifold chart with a union of coordinate subspaces).*

Proof. Away from the stated singularities, X is a smooth manifold and is locally smoothly contractible. At a weighted-conical singularity use Corollary 2. A union of coordinate subspaces is invariant under ordinary radial dilation, so normal-crossing charts are the weight-one case. □

For example, the cusp $y^2 = x^3$ contracts by $(x,y) \mapsto (t^2 x, t^3 y)$, while a node or a union of coordinate subspaces contracts by ordinary radial scaling. These singular examples therefore do not create extra local smooth singular cohomology.

4 Why this does not settle the general case

The source's local contraction is Lipschitz and semialgebraic. Its composition with an arbitrary ambient-smooth simplex need not be smooth: the simplex can meet the contraction's nonsmooth strata along a complicated closed set. A flattened semialgebraic triangulation can make prescribed semialgebraic simplices smooth at their faces, but injectivity would require a relative smoothing theorem fixing an arbitrary smooth boundary cycle. Neither a C^1 semialgebraic triangulation nor semialgebraic-current homology supplies that step.

Likewise, a Whitney algebraic set has a conically smooth stratified structure, but a conically smooth contraction need not be smooth for the ambient-subspace predifferentiable structure used in the question. Thus the argument above is a genuine partial result: it proves the desired map is an isomorphism on a substantial explicit class while leaving the arbitrary algebraic germ open.

Source and verification note

The exact open sentence occurs after Diagram (7) in Igor Baskov, *The de Rham cohomology of soft function algebras*, arXiv:2208.11431. The literature sweep also checked arXiv:1505.03970 (semialgebraic C^1 triangulations), arXiv:1404.4796 (semialgebraic current homology), and arXiv:2202.00131 (diffeological smooth singular complexes); none supplies the missing relative ambient-smooth comparison.